

Intersubject Correlation of Physiological Responses to Discrete Emotional Videos

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INTRODUCTION

- Intersubject correlation (ISC): An analysis technique to measure response synchrony (Hasson et al. 2004).
- Emotional states, represented by hedonic valence and high arousal, make brain activity and autonomic responding more synchronized: increased ISC indices (Nummenmaa et al., 2012, Golland et al., 2014).
- However, understanding how physiological synchrony varies depending on distinct emotions, such as fear, is poorly developed.
- To this aim, we analyzed how four emotions elicited by video clips influence physiological synchrony in seven physiological measurements.

METHODS

- The CASE (continuous annotation signals of emotions) dataset included 30 participants' physiological signals in response to 8 videos; each pair aimed to elicit amusement, boredom, relaxation, and fear (Sharma et al., 2019).
- Seven physiological features: heart rate; HR, galvanic skin response; GSR, skin temperature; ST, respiration rate; RR, the signals of electromyographic zygomaticus major; EMG-z, corrugator supercilli; EMG-c, and trapezius; EMG-t
- ISCs were computed for all physiological responses to each video (Figure 1). The ISC dataset was analyzed using repeated-measures ANOVAs to find the effect of emotions on physiological synchrony.

RESULTS

- There were significant effects of emotions on HR, GSR, RR, ST, EMG-z, and EMG-t physiological synchrony. Bonferroni comparison results showed that ISCs responding to fear were consistently higher than other emotions.
- Physiological synchrony in response to other emotions depended on each physiological measure (In detail, refer to Figure 2).

CONCLUSION

- We found that emotional videos significantly affected physiological synchrony across measurements.
- High physiological synchrony in response to fearful videos was observed consistently across physiological measures. This finding is compatible with former studies (Nummenmaa et al., 2012, Golland et al., 2014), showing that hedonic valence and high arousal are associated with increased brain synchrony.
- ISCs, as elicited by other emotions, depended on physiological measurements. These results should be investigated in future research.

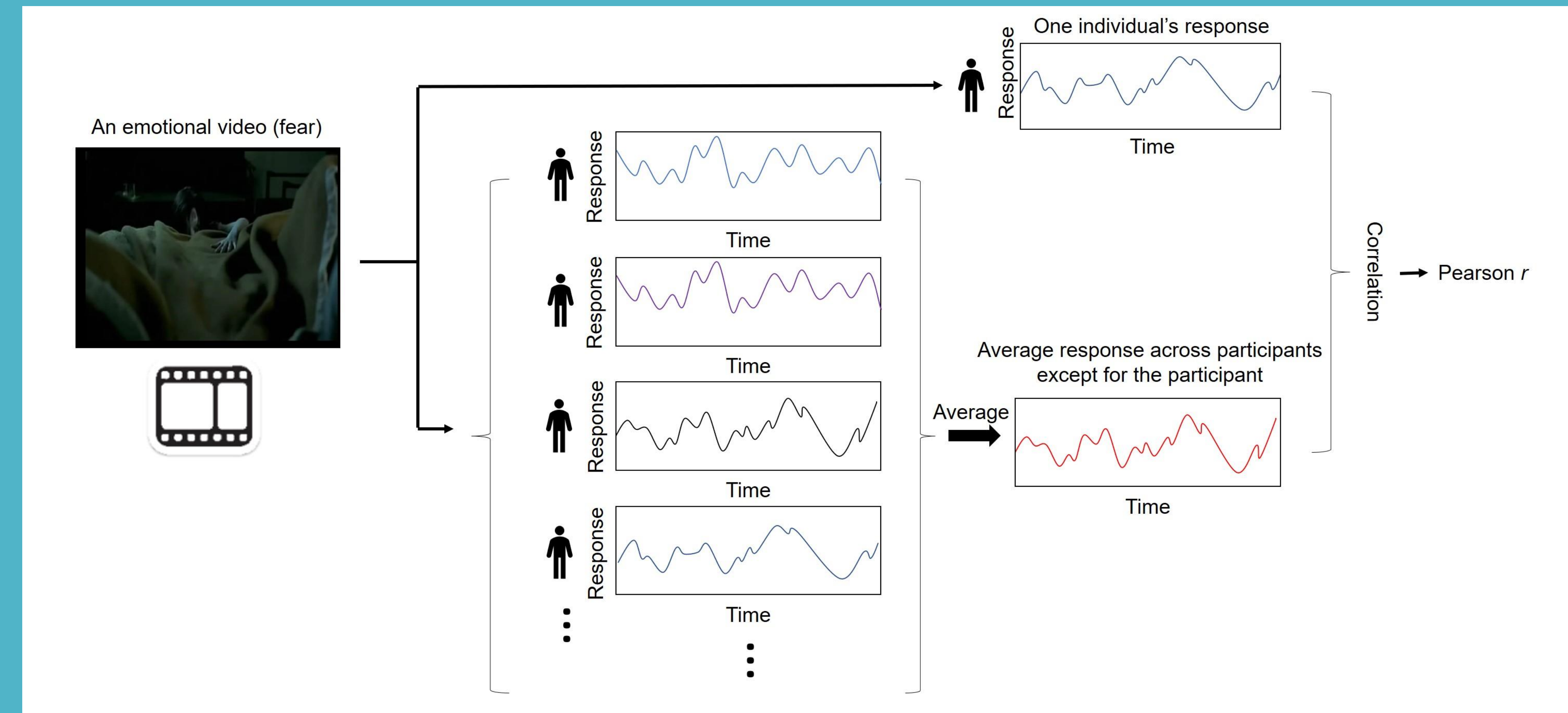


Figure 1. The intersubject correlation analysis procedure. One participant's temporal response was correlated with the average response values derived from the other participants. The computed Pearson correlation (r) represents the individual participant's ISC indices, namely response synchrony in relation to other participants.

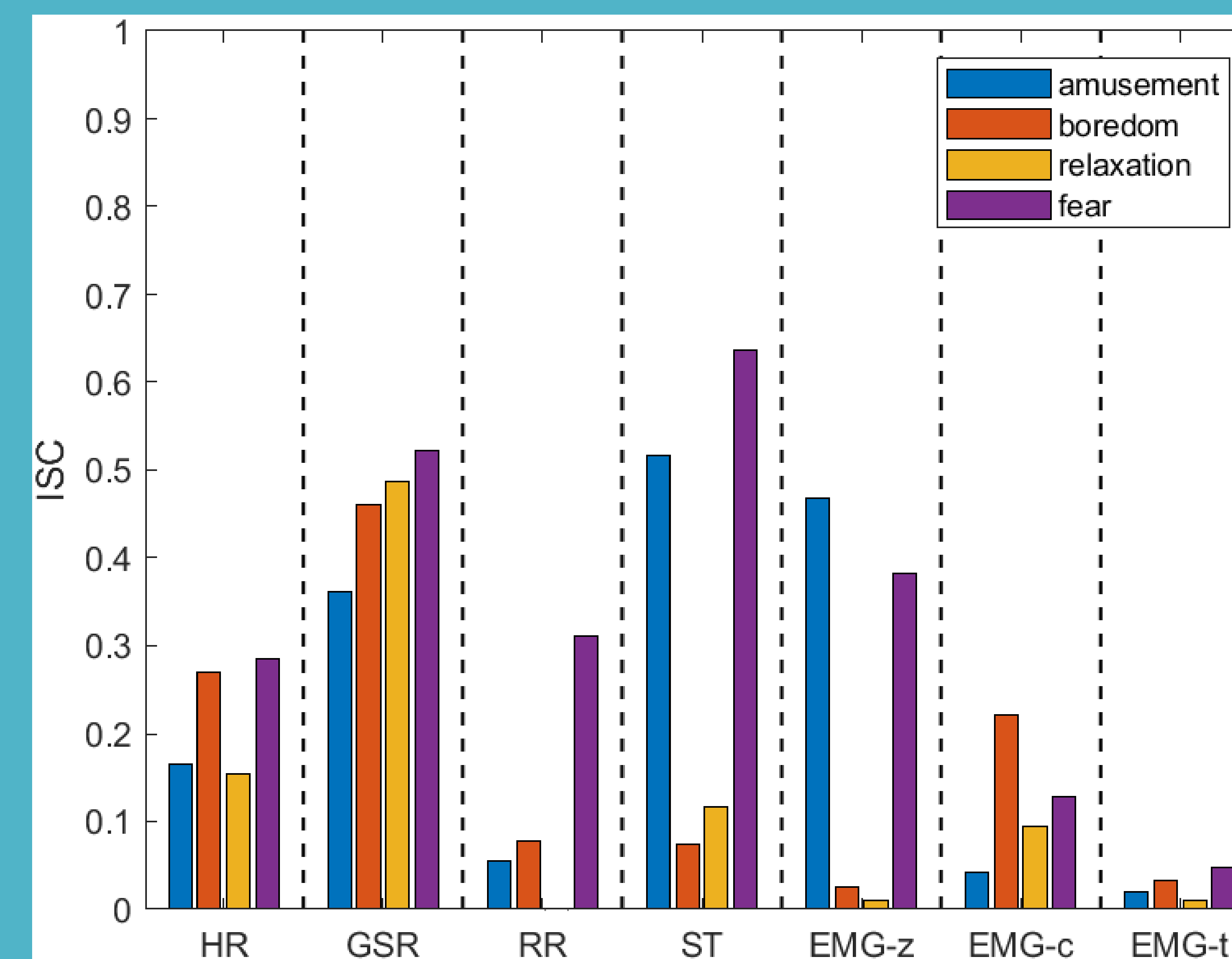


Figure 2. The results of repeated-measures ANOVAs as follows:

- Except for EMG-t, there were main effects of emotion on physiological synchrony.
- ISCs in response to fear were higher than other emotions in most physiological measurements.
- Response synchrony with amusement was high in ST and EMG-z signals but low ISCs in HR, GSR, RR, and EMG-c.
- Boredom made participants' HR and EMG-c responses more synchronized but did not in RR, ST, or EMG-z measures.
- The relaxation ISCs were consistently low, apart from GSR.

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